



Your PC without over clocking

It's easy to overclock, but not everyone can...

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Wouldn't it be great if you could speed up your PC without having to overclock your hardware, and risk damaging it all? With a few tweaks in the motherboard BIOS, you can speed up boot times as well as overall system performance. BIOS settings are easy to change, but the naming conventions used in them may vary from one board manufacturer to another. To boot into your BIOS, press [Delete], [F1] or [F2] keys on your keyboard while the PC boots. Once you've made the changes, save the BIOS settings and reboot.

Note: In case of any problems with booting the system after changing settings, you can clear the CMOS settings by switching jumper positions on your motherboard or by pressing a reset CMOS

switch. Refer to the motherboard manual for the position of this switch or jumper.

CPU and memory

Most modern motherboards are designed to slow down CPUs to keep temperatures in control and also to optimise power consumption. Use the CPU configuration menu and set the CPU multiplier and board bus speed as stated by the manufacturer. Be sure to enter accurate values. Any mistakes here, would overclock or underclock the processor. For example, a Core i7 965 runs at 3200 MHz (24 x 133). Similarly, in the memory configuration menu, set the memory speed and timings as per the specifications. If timings aren't available, set just the speed.

Disable CPU throttling

Features like EIST (better known as SpeedStep) from Intel and Cool 'n' Quiet from AMD are designed to lower speeds

when the processor isn't in use. If you want the maximum possible performance all the time, turn off these features under the Advanced Settings option in the BIOS. Just keep in mind that the processor might run hotter than usual.

Boot orders

Selecting the right boot order of drives reduces boot times by a few seconds. Make sure you have no USB devices when you are booting your system. An option to disable USB boot devices is also present. Set the bootable hard drive to be the first drive to boot. Remember that you will have to set your CD/DVD drive as bootable if you want to install an operating system. Your primary boot drive should always be the drive to boot.

Enable Quick Boot

The Quick Boot feature boots the system without any of the splash screens from your motherboard appearing. Similarly, also disable any RAID controllers and network boot options. This saves a couple of seconds during the bootup.

Auto-detection of IDE drives

If you are still using IDE devices, the board by default, detects all the drives during bootup. There are a few valuable seconds that can be saved here. Boot into the BIOS and select the drive channels the drives are connected to, and manually enter the details for them. Disable the auto detection feature.

Integrated graphics memory

Motherboards with integrated graphics chipsets require a decent amount of memory to perform well. At the same time, too much memory should not be allotted to it. 128MB to 256MB should be sufficient for most integrated graphics. This would reduce the amount of system memory your operating system has. If you use a discrete graphics solution, disable the allocated memory completely if possible.

Boot loader timings

If you have two or more operating systems installed, you are bound to have a boot loader installed. The boot loader window appears on screen for 10 seconds by default. Set the operating system you use most as default, and reduce this time. 